For Examiner's Use

4 **(a)** A narrow beam of light of wavelength 632 nm is incident normally on a diffraction grating as shown in Fig. 4.1.

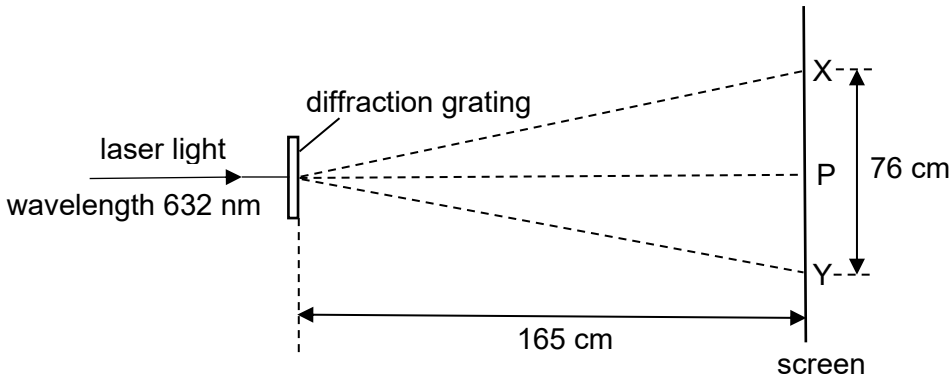

Fig. 4.1

Spots of light are observed on a screen placed parallel to the grating. The distance between the grating and the screen is 165 cm.

The brightest spot is P. The spots formed closest to P and on each side of P are X and Y. X and Y are separated by a distance of 76 cm.

(i) Calculate the number of lines per metre on the grating.

number per metre = ……. [3]

(ii) Light of wavelengths 632 nm and 638 nm is now incident normally on the grating. Two lines are observed in the first order spectrum and two lines are observed in the second order spectrum, corresponding to the two wavelengths.

State two differences between the first order spectrum and the second order spectrum.

1. ……..……..……

……...

2. ……..……..……

……... [2]

For Examiner's Use

(b) **(i)** The grating in **(a)** is now rotated about an axis parallel to the incident light, as shown in Fig. 4.2.

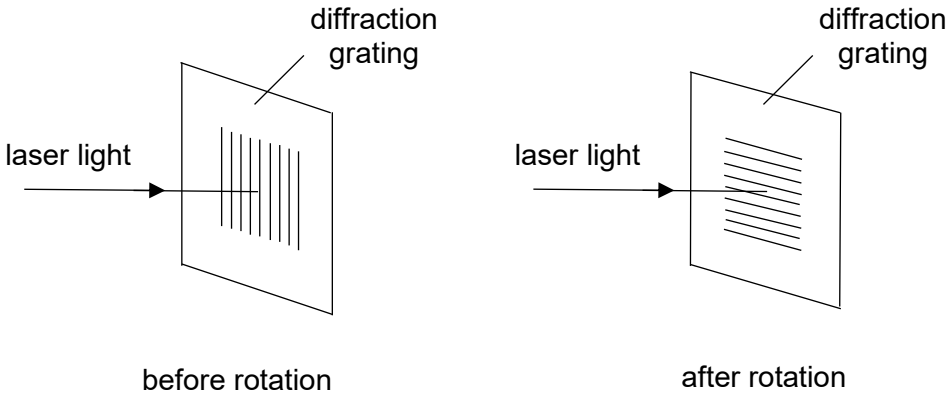

Fig. 4.2

State what effect, if any, this rotation will have on the positions of the spots P, X and Y.

……

…… [2]

(ii) In another experiment using the apparatus in **(a)**, it was noticed that the distances XP and PY, as shown in Fig. 4.1, are not equal.

Suggest a reason for this difference.

…… [1]